

Sleep Impact Over Blood Pressure: An Analysis of the NHANES Dataset

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Introduction

Sleep on average represents one-third of our life and is considered a necessity for human health and wellbeing. It affects nearly every system in the body, influencing brain function, physical health, emotional well-being, and metabolism. This short data analysis was performed to verify the impact of the amount of sleep on the wellbeing of a population by testing for a correlation between self-reported sleep duration and average clinical blood pressure.

Data sample

The analysis utilizes the 2021-2023 National Health and Nutrition Examination Survey (NHANES) cycle [1]. The specific files used include:

- **DEMO_L.XPT (Demographics):** Provided the unique respondent sequence number (SEQN) and participant age (RIDAGEYR).
- **SLQ_L.XPT (Sleep Disorders):** Provided the reported hours of nighttime sleep (SLD013).
- **BPXO_L.XPT (Oscillometric Blood Pressure):** Provided three consecutive clinical systolic blood pressure readings (BPXOSY1, BPXOSY2, BPXOSY3).

The blood pressure data was normalized using Python's pandas package by calculating the mean and standard deviation of the dataset to generate Z-scores. Furthermore, `pandas` was used to calculate the Pearson correlation matrix between the variables, and the `seaborn` library was utilized to generate the corresponding heatmap visualizations.

Results and discussion

The analysis visually and statistically confirms a correlation between age and blood pressure. As shown in the calculated Pearson Correlation matrix (Figure 1), there is a positive correlation of 0.424 between age and mean systolic blood pressure, indicating age plays a crucial role.

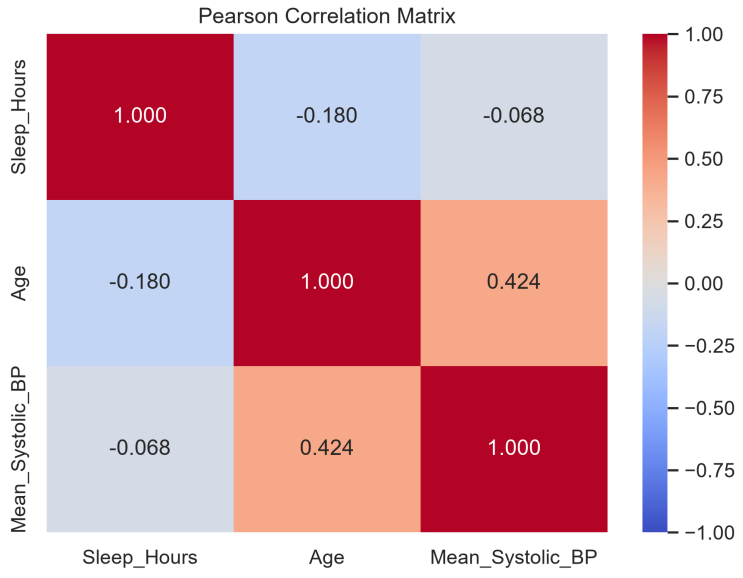


Figure 1: Pearson Correlation Matrix displaying relationships between Sleep Hours, Age, and Mean Systolic BP.

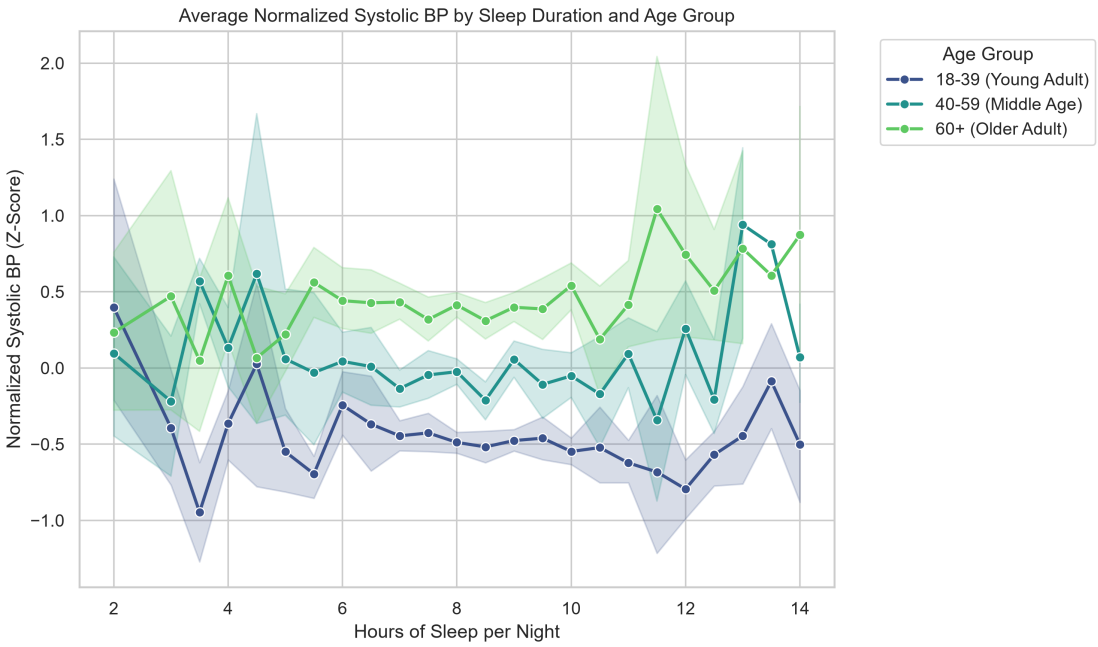


Figure 2: Average Normalized Systolic BP by Sleep Duration and Age Group.

However, sleeplessness is not statistically proven to affect blood pressure in a measurable way within this dataset. The Pearson correlation between sleep hours and mean systolic blood pressure is exceptionally weak at -0.068 (near 0), indicating no conclusive correlation. The low confidence intervals visible on the line chart at the bellow 6hrs sleep and over 10 hrs sleep are due to the high variance and low sample numbers for the individuals within that "sleep time range". Limitations of this conclusion include the reliance on questionnaire-based data collection for sleep hours, a lack of data on chronically "sleepless" individuals and not taking into account over relevant factors, such as medications, diets or illnesses, which are all also availble within the NHANES dataset [1].

References

[1] Centers for Disease Control and Prevention. *National Health and Nutrition Examination Survey data 2021-2023*. National Center for Health Statistics. Available at: https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?Cycle=2021-2023.